

A Prospective Controlled Study in the Prevalence of Posttraumatic Headache Following Mild Traumatic Brain Injury

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ABSTRACT

Objective. To establish the prevalence of post traumatic headache, persisting at 3 months following minor traumatic brain injury.

Design. A prospective controlled study of patients admitted with a diagnosis of mild traumatic brain injury and matched orthopedic controls over 12 months during 2004.

Setting. A level two inner city Emergency Department in Sydney, Australia.

Patients. One hundred eligible sequential admissions with mild traumatic brain injury as defined by American Congress of Rehabilitation Medicine, 1993, and 100 matched minor injury controls with nondeceleration injuries.

Interventions. Subjects were part of a study on prediction of postconcussive syndrome and had neuropsychological tests, balance test and pain recordings taken at the time of injury, at 1 month and at 3 months post injury.

Outcome Measures. Main measures were the reporting of headache "worse than prior to the injury" and concordant with the definition of Posttraumatic Headache according to International Headache Society Classification of Headache Disorders 2003.

Results. 15.34% of those with minor head injury continued to complain of perisistant posttraumatic headache at 3 months compared to 2.2% of the minor injury controls.

Conclusions. To the authors' knowledge this is the first controlled prospective study in the prevalence of posttraumatic headache following mild traumatic brain injury.

Key Words. Mild Traumatic Brain Injury; Posttraumatic Headache; Epidemiology; Pain; Prevalence; Head Injury

Introduction

Current literature in posttraumatic headache (PTH) following closed head injury indicates a variation in prevalence from 30–50% [1]. In Carroll et al.'s World Health Organization Neurotrauma Taskforce study into mild head injury 2003 [2], it is indicated that this variation is largely because of differences in definitions and method-

ologies of the studies. In both Wade et al.'s 1998 [3], and van der Naalt et al.'s 1999 [4] longitudinal studies, mild traumatic brain injury (MTBI) is grouped together with moderate head injury. Many other studies are retrospective and have a significant selection bias. In Ingebrigtsen et al.'s [5] prospective noncontrolled study of 100 consecutive patients suffering MTBI in Scandinavia, 42% reported PTH at 3 months and over 60% reported the presence of one or more symptoms. Forty percent fulfilled the criteria for postconcussive disorder (PCD). Indeed, Kraus [6] cites this as a reason for initiating her group's 2005 prospective controlled longitudinal study of MTBI. In that

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US-based study she notes a prevalence of 79% of PTH at 3 months post injury and 36% at 6 months.

In Mickeviciene et al.'s [7] large retrospective study of motor accident victims undertaken in 2002 in Lithuania, a number of methodological aspects may have confounded the very low result of PTH (less than 11%). Mickeviciene et al. suggest that the low result is related to a lack of a third party insurance scheme. They refer to the inflated incidence of PTH and postconcussive syndrome (PCS) in Western countries as a reflection of the compensation system. The methodological characteristics of Mickeviciene et al.'s study include a recall bias (patient were posted surveys up to 36 months post motor vehicle accident (MVA), its retrospective nature and the lack of definitions for PTH and MTBI. Mickeviciene et al. then undertook a prospective study in 2004 [8] and found 11.5% complained of PTH for more than 3 months post injury and that this figure was significantly higher than the control group. However, by 1 year, they report no difference in the incidence of headache between the two groups. In this study, they defined headache as a symptom of the PCS complex, as "headache on more than seven days per month."

For the prevalence of PTH following MTBI to be accurately estimated, rigorously applied definitions need to accompany a prospective controlled study. Definitions of MTBI and PCS should also be established prior to data collection. Definitions used in this prospective study are referred to in the next section (entitled Definitions).

Certainly, there are a number of studies such as Barnat's [9] that identify the chronic PTH population as having a number of characteristic demographic features such as high school education only. Others have identified acute stress reactions at the time of injury [10] and low satisfaction with current occupation [11] as characteristic features.

In Australia, Ponsford et al. (2000) [12] suggested that 24% of adults who suffered MTBI complained of persistent symptoms at 3 months post injury. Although PTH was not a specific outcome measure, headache was the most common symptom noted. Ponsford et al. [12] identified a number of demographic characteristics as being more common in the group that suffered persistent symptoms, these included; a previous head injury, previous psychiatric or neurological conditions, being female or sustaining the injury in an MVA.

All studies point to the need to undertake a prospective controlled study in PTH following

MTBI using stringently applied definitions and time frames. Carroll et al. [2] emphasize the importance of collecting data on pre-morbid conditions, personality type and postinjury distress, depression and pain.

At St Vincent's Hospital in Sydney, Australia, our research group [13] have completed a controlled prospective study in the prediction of PCS using a number of simple bedside tests in the emergency department (ED). One of the tests used was a simple visual analog scale (VAS) of pain. The following study reports the prevalence of PTH in both the MTBI cohort and the minor injury control group.

Definitions

The definitions used in the study were as follows:

Posttraumatic headache following mild head injury is defined by Second Edition of the International Headache Society Classification of Headache Disorders 2003 [14] as a headache that develops within 7 days after head trauma or regaining consciousness following head trauma. The head trauma must have all the following characteristics:

1. Either no loss of consciousness or loss of consciousness for less than 30 minutes.
2. GCS score of equal or greater than 13.
3. Symptoms or signs diagnostic of concussion.

If the headache persists for longer than 3 months, it is described as chronic. It may occur with or without abnormalities in neuropsychiatric testing, electroencephalogram, neuroimaging, cerebrospinal fluid studies, evoked potentials or vestibular studies.

Postconcussive syndrome or PCD is defined by the Diagnostic and Statistical Manual Four Edition Text Revision [15] as a clinical state in which head trauma, as evidenced by concussion and cognitive deficits, is accompanied by three or more postconcussion symptoms persisting for more than 3 months. Such symptoms can include somatic complaints (such as headache, dizziness, nausea, fatigue, sleep disturbance, restlessness, light or noise sensitivity), cognitive deficits (such as poor concentration, memory problems, slowed thinking), and affective problems (such as depression, frustration, irritability).

Mild traumatic brain injury is defined by both Alexander [16] and the American Congress of Rehabilitation Medicine (1993) [17] as a traumatically induced disruption of brain function charac-

terised by a brief loss of consciousness (30 minutes or less), any alteration in mental state, anterograde or retrograde amnesia, normal computed tomography (CT) scan and posttraumatic amnesia (PTA) less than 24 hours.

Carroll et al. [2] in a noted exception includes abnormal CT scans and a number of complex qualifications in the definition but as this research took place in an active ED setting, the definition of Alexander et al. [16] and the American Congress of Rehabilitation Medicine 1993 [17] was accepted for this controlled study.

Carroll et al.'s WHO Neurotrauma Taskforce in MTBI [18] proposed a definition of mild traumatic head injury, and included "an intracranial lesion not requiring surgery" but put a large number of limitations on the definition stating "These manifestations of MTBI must not be due to drugs, alcohol, medications, caused by other injuries or treatment for other injuries (e.g. systemic injuries, facial injuries or intubations), caused by other problems (e.g. psychological trauma, language barrier or coexisting medical conditions) or caused by penetrating craniocerebral injury." According to Ponsford [19], Carroll's definition highlights the complex nature of this condition, including the numerous factors which potentially complicate initial diagnosis and treatment. These further qualifications to the definition make it somewhat unwieldy for use in the clinical or research setting of an ED where according to a pilot study undertaken by our group Sheedy et al. [13] as many as 55% of MTBI patients have positive blood alcohol readings.

Methodology

This was a prospective controlled study of 100 patients who were admitted to the ED of St Vincent's Hospital, Sydney, in 2004 with the diagnosis of MTBI. The MTBI group was compared with 100 patients with low impact orthopedic injuries such as falls in which fractures or soft-tissue injuries were sustained (minor injury group). The methodology for this study is also described in detail in the article entitled ED assessment of MTBI and prediction of postconcussion syndrome by Sheedy et al. (submitted) which is further to a pilot study examining prediction of postconcussive symptoms at 1 month by Sheedy et al. in 2006 [13].

The above two groups were followed for 3 months to determine if a bedside recording of pain, balance and neurocognitive tests may determine who was at risk of developing PCS. Ethics

approval was obtained and all patients who met eligibility criteria and who were consented accordingly were included in the study.

Participants

MTBI Subjects

Inclusion Criteria

Patients who attended St Vincent's Hospital ED with a closed head injury and who met the following criteria were recruited to the study:

1. Seventeen years or over.
2. Fluent English speakers.
3. GCS 13–15 in ED.
4. A negative head CT scan (no intracranial pathology).
5. Any of the following indicators of concussion: loss of balance, altered consciousness, retrograde or anterograde amnesia, disorientation, confusion, vomiting, nausea, blurred vision or headache.

Exclusion Criteria

Patients with closed head injury and any of the following were excluded from the study:

1. GCS < 13 in ED.
2. Positive head or cervical spine CT scan.
3. Any new or old intracranial pathology.
4. Skull fractures.
5. Concurrent injuries with Abbreviated Injury Score [20] equal to or below two.
6. Hearing disorder precluding completion of assessment.

Minor Injury Control Group

Some authors including Evans [21], Warner [22] Packard, [23] and Carroll [2] have suggested that PTH may be part of a psychological or stress reaction to trauma. Therefore, the pain scores of the MTBI group in this current study was compared with that of patients with minor orthopaedic or soft-tissue injuries sustained through nondeceleration mechanisms. This was to explore the association of PTH to MTBI while controlling for the psychological effects of trauma (e.g., distress, stress or depression). This particular control group were chosen as their injury site was well away from the head and neck (e.g., lower or upper limbs). The inclusion of minor injury controls who had not suffered associated deceleration injuries was critical because of reports that those suffering limb or torso fractures may also sustain an MTBI, although it may not be the focus of treatment or assessment. These patients were identified from

the St Vincent's Hospital Emergency Department Information System (EDIS) and had injuries with an Abbreviated Injury Score of two or less. They needed to be over the age of 17 and be able to speak English fluently.

Measures

Pain Testing

Visual analog scores of pain (range 0–10) were presented to both MTBI patients as well as minor injury controls. Both groups were asked by researchers to accept zero as a measure of no pain and 10 as the worst pain imaginable in accordance with accepted measures of pain in an ED, described by Todd et al. [24]. MTBI patients were asked to rate their "headache," while minor injury controls were asked about "any pain" which could have occurred in a variety of locations. All subjects were also asked to rate any other symptoms they were experiencing. Data on headache alone were not collected in the minor injury control group.

Postconcussive Symptoms

At one and 3 months following injury, postconcussive symptoms were measured on the Rivermead Post Concussion Symptoms Questionnaire [25]. The questionnaire was administered by telephone to both groups.

The Rivermead Post Concussion Symptoms Questionnaire [25] includes a question which specifies: "After a head injury or accident some people experience symptoms which can cause worry or nuisance. We would like to know if you now suffer any of these symptoms. As many of these symptoms occur normally, we would like to compare yourself now, with before the accident. For each one, please tell us the number closest to your answer:

- 0—not experienced at all
- 1—no more a problem now than before
- 2—a mild problem
- 3—a moderate problem
- 4—a severe problem.

Compared with before the accident do you now (in the last 24 hours) suffer from headaches?"

Respondents were then asked to indicate a score ranging from zero to four. The questionnaire goes on to ask about 15 other symptoms and then asks respondents to add any symptoms not on the checklist. This questionnaire has been validated and assessed for reliability by King et al. for the diagnosis of PCS [25]. As the data were collected through interview, subjects were able to be clarified scoring details, at the time, with the interviewer.

Alcohol

As it was expected that a significant proportion of subjects may be affected by alcohol, and as alcohol intoxication would be likely to affect performance on tests, participants were breathalyzed with a "Lion 300 Alcometer" or alcohol detector following completion of the testing battery. Participants were also asked to estimate the number of drinks they had consumed over the past 24 hours.

Neurocognitive Testing and Balance Testing

As this study was part of a larger study on the prediction and prevalence of PCS, a number of other measures were collected from both groups at the same time including measures for PTA; *the GOAT—Galveston Orientation and Amnesia Test* [26] and *the modified Westmead Post Traumatic Amnesia Scale* [27], and neurocognitive functioning; *the modified Rapid Screen of Concussion* [28]. Further, at one and 3 months, postconcussive symptoms were measured on the Rivermead Post Concussion Symptoms Questionnaire [25].

Patients with MTBI were also asked to undertake a postural stability test—the modified Balance Error Scoring System according to Gruskiewicz [29]. The minor injury group did not perform this test for practical reasons as many of this group had injured their lower limbs. This testing was done to determine if impaired balance was a predictor of PCS.

It is important to note that testing of the subjects took approximately 20 minutes. The results of the cognitive testing and regression analysis regarding the prediction of postconcussion syndrome are presented in Sheedy et al. (submitted). Sheedy's article does not examine the issue of PTH but rather discusses the epidemiology of PCS.

Procedure

Data Collection

Participants were identified via the EDIS which alerted the researchers to a case that might be appropriate for the study. Prior to approaching the patient, the medical officer involved was consulted to ensure that patient recruitment was safe (e.g., the patient was medically cleared to mobilize). Data were collected 7 days a week during working hours. Before participation, patients received an information sheet explaining the purpose, procedure and requirements of the project. As the MTBI participants may have been affected by PTA or by drugs or alcohol, verbal consent was sought again at the time of the follow-up phone call.

Acute Testing Protocol—MTBI Subjects

Initial information was obtained from the medical record including age, medical history, imaging results, neurological observations, and blood alcohol level. Participants were questioned about mechanism of injury, previous head injuries, previous contact with a psychiatrist or psychologist, occupation, occupational satisfaction, years and level of education and the amount of drugs/alcohol consumed prior to their injury. Participants then completed the entire assessment in the following order: GOAT, modified Westmead PTA scale, RSC, balance assessments and then pain (VAS) and other acute symptoms. All participants were breathalyzed at the completion of the assessment.

Acute Testing Protocol—Minor Injury Controls

Participants in the minor injury comparison group completed the same battery of tests apart from the balance testing. Practical reasons such as limb injury made this test unable to be undertaken. The minor injury control group was age- and gender-matched to the MTBI group.

Follow-up Testing Protocol—I Month and 3 Months Post Injury

Follow-up testing was completed by telephone interview at 1 month and 3 months post injury. Both MTBI and minor injury control participants were followed up by telephone as it has been reported by Dikmen et al. [30] that injuries other than MTBI can affect reported post “concussive” symptoms. As noted above, question one of the Rivermead Post Concussion Symptoms Questionnaire [25] focuses on headache and was the sole method of determining the incidence of PTH. Although patients had been shown a VAS when admitted to the ED, they did not have the scale in their possession at the time of the phone call and a VAS was not recorded.

Results

Group Differences

Descriptive statistics were studied and group differences (MTBI vs minor injury) on the following variables were examined: age, alcohol levels, and years of education. Chi-squared tests were used for categorical variables and t-tests or its nonparametric equivalent, the Mann-Whitney U test, were used for continuous variables (age, years education, blood alcohol level) according to distribution.

Demographic Variables

Gender

The gender proportions did not differ significantly across groups (MTBI: 78% male; minor injury controls: 77% male) ($\chi^2(1, N = 200) 0.03, P > 0.05$).

Age

The mean ages for the MTBI and minor injury groups were compared. The mean age of the MTBI group was 33.64 years and did not differ significantly from the minor injury group (mean age 32.15 years) ($t(0.62), P > 0.05, N = 200$).

Education

The minor injury control group had on the average completed significantly more years of education than had the MTBI group (14.26 vs 13.19; $t(-2.59), P < 0.01, N = 200$). Significantly more of the minor injury group participants had been educated to tertiary level (65% compared with 48% of the MTBI group) ($\chi^2(2, N = 200) 7.89, P < 0.05$).

Alcohol

A significantly greater proportion of the MTBI group had a positive blood alcohol reading (20%) compared with the minor injury group (8%) ($\chi^2(1, N = 200) 8.30, P < 0.05$). The mean breath alcohol level (BAL) of the MTBI group was 0.013. This was significantly higher than the level of the minor injury group (mean BAL 0.004 U(4289.50), $P < 0.05, N = 198$). Of those in the MTBI group 20% had breath alcohol levels between 0.006 and 0.174, and 78% had no recorded alcohol on their breath (2% of missing data). In the minor injury controls 8% had breath alcohol recordings of between 0.014 and 0.074 and 92% had no recorded alcohol on their breath. It should be noted that median time from injury to breathalyzer testing was 540.00 minutes (Interquartile range 788.00 minutes) in the MTBI population while in the control minor injury group the median injury to breathalyzer time was 437.50 minutes (Interquartile range 915.00 minutes).

Previous Head Injuries

The median number of reported previous concussions was 2.00 in both groups; however, a significantly greater proportion of the MTBI group compared with the minor injury group reported having sustained a previous head injury (55% compared with 37%) ($\chi^2(1, N = 200) 6.52, P < 0.05$). Group means for age, education, blood alcohol level, previous head injuries and significant group differences can be seen in Table 1.

Table 1 Differences in age, education, blood alcohol, and number of previous head injuries between MTBI and the minor injury controls

	MTBI N = 100	Minor Injury N = 100	
	Mean (SD)	Mean (SD)	P Value
Age	33.64 (12.67)	32.15 (10.21)	0.36
Education (years)	13.19 (3.32)	14.26 (2.46)	0.01
Blood alcohol level	0.01 (0.04)	0.00 (0.01)	0.02
Median number previous head injuries	2.00	2.00	—

MTBI = mild traumatic brain injury.

Mechanism of Injury

The majority of the MTBI group suffered an alleged assault (40%). A further 24% sustained an MTBI during a low fall while 23% were injured either as a result of a collision with an object or were struck by an object. The remainder, 12% of the MTBI group, were involved in a road traffic accident, which included those who were pedestrians and cyclists struck by a vehicle. By comparison, the majority of the minor injury group were injured as a result of a low fall (53%) or suffered a laceration or crushing injury (28%). Eleven percent of this group sustained their injury as a result of a collision with an object, 3% had been involved in an alleged assault and 2% had been injured in road accidents. These category differences across groups were statistically significant ($\chi^2(5, N = 200) 86.06, P < 0.001$). A summary of the mechanism of injury for the MTBI and minor injury groups can be seen in Table 2.

Occupation

Seventy-five percent of the MTBI group was employed compared with 87% of the minor injury group. The proportion of each group who were students was similar between the two groups (11% MTBI compared with 9% minor injury group). More of the MTBI group were either unem-

Table 2 Mechanism of injury: MTBI compared with minor injury group

	MTBI N = 100	Minor Injury N = 100
	% Injured	% Injured
Alleged assault	40	3
Low fall	24	53
Collision with/struck by object	23	11
Laceration/crush injury	N/A	28
Road accident	13	2
Other	0	3

MTBI = mild traumatic brain injury; N/A = not available.

Table 3 Differences in occupational groups between MTBI and minor injury group

	MTBI N = 100	Minor Injury N = 100
	%	%
Employed	75	87
Unemployed	6	1
Home duties/retired/travelling	8	3
Student	11	9

MTBI = mild traumatic brain injury.

ployed, retired, on home duties or travelling (14%) compared with the minor injury group (4%). These differences were not statistically significant ($\chi^2(3, N = 200) 6.93, P > 0.05$). Occupational details can be seen in Table 3.

Time from Injury to Assessment

Median time from presentation at the ED to testing was 9 hours in the MTBI group and 7.29 hours in the minor injury group. There were, however, large variations in both groups from time to injury to time to assessment (interquartile range: 13.13 hours MTBI, 15.25 hours minor injury). These differences were not significant $U(4604.00), P > 0.05, N = 200$). Reasons for delay to testing were numerous but included waiting for medical clearance for the patient to be mobilized (required for balance testing), delays to ED presentation and waiting for imaging and other necessary investigations to be completed. Patients were frequently assessed just prior to discharge from the ED.

Pain Measurement

In terms of pain measurement, the following graph depicts the VAS scores of pain at initial injury. The control group is reporting pain at any site in their body while the MTBI group are reporting headache only. Median reported pain score was 4.00 in both groups (interquartile range of 5.00 in the MTBI group compared with 4.00 in the minor injury group). These differences were not statistically different $U(4185.50), P > 0.05, N = 199$ (Figure 1).

At 1 month and 3 months, subjects were asked to rate their headache using question one of the Rivermead Postconcussive Questionnaire. Results are depicted both in graphical form and in tables (Figures 2 and 3; Tables 4 and 5).

The natural history of PTH in this cohort of people who have suffered MTBI demonstrates an approximate 50% reduction in reported incidence

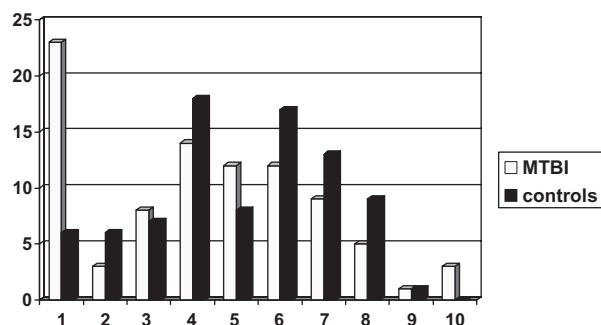


Figure 1 VAS scores of pain at time of injury. X axis indicates the VAS scores from 0—no pain to 10—worst pain imaginable. Y axis percentages of total study population (light color) and minor injury control population (dark color). MTBI group refers to pain at site of injury—headache. Control group refers to pain at location of injury—variety of locations distant from head, not headache. VAS = visual analog scale; MTBI = mild traumatic brain injury.

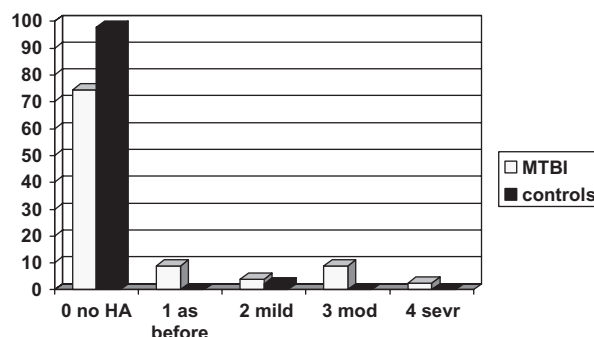


Figure 3 Headache scores at 3 months. X axis indicates the headache scores from the Rivermead Postconcussive Questionnaire. 0 = no headache; 1 = I do have a headache but it is no more of a problem now than before my injury; 2 = mild headache; 3 = moderate headache; 4 = severe headache. Y axis represents percentage of MTBI study population (light color) and minor injury control population (dark color). HA = headache; MTBI = mild traumatic brain injury.

between the first and third month. That is to say at 1 month 30.47% suffered PTH but by 3 months only 15.35% still complained of headache. A concomitant rise in the number of subjects reporting no headache from 58.57% to 86.17% can also be seen (Figure 4; Table 6).

In our prospective study, the incidence of headache at the time of injury was 100% and at 1 month it was 30.4%, while at 3 months it was 15.35%.

The incidence of PTH in the control group was significantly lower without diminution over time 2.12% at 1 month and 2.25% at 3 months.

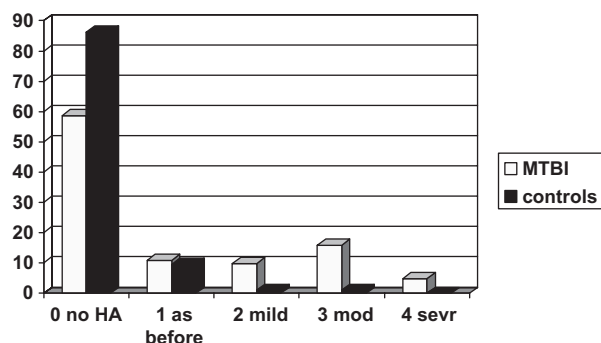


Figure 2 Headache scores at 1 month. X axis indicates the headache scores from the Rivermead Postconcussive Questionnaire. 0 = no headache; 1 = I do have a headache but it is no more of a problem now than before my injury; 2 = mild headache; 3 = moderate headache; 4 = severe headache. Y axis percentage of study population (light color) and minor injury control population (dark color). HA = headache; MTBI = mild traumatic brain injury.

There was a correlation of VAS scores for headache at the time of injury and scores on the Rivermead Post Concussion Symptoms Questionnaire, suggestive of PCS which was significant at 0.331 (P value 0.01).

Discussion

In this study, 15.35 % of people who suffered an MTBI complained of PTH 3 months post injury. This contrasts with 2.25% of those in the minor injury control group who reported PTH. In Carroll et al.'s WHO task force on MTBI [3], they indicate that most self reported symptoms are resolved within 3 to 12 months following head injury. Our results demonstrate that a significant proportion suffer with PTH for more than 3 months, although our study was not able to follow up patients with PTH for 12 months by which time Carroll suggests that most of the PTH would have resolved.

It is relevant to note that having PTH for three to 12 months may result in significant absenteeism from work, maladaptive chronic pain behaviors and loss of quality of life. As such, it represents a significant public health issue particularly when one considers that MTBI is a common daily presentation to the ED. British epidemiological studies have estimated 250–300 hospital admissions involving head injury per 100,000 of the population [31] and 90% of these are minor head injuries [32]. This is confirmed by Carroll et al. [2] in the WHO Neurotrauma Task force on MTBI

Table 4 The table below indicates raw numbers and percentages in brackets

Scores	0 No Headache	1 as Before	2 Mild	3 Moderate	4 Severe	Lost to Follow-up
MTBI	48 (58.53%)	9 (10.97%)	8 (9.75%)	13 (15.85%)	4 (4.87%)	18
Minor injury	81 (86.17%)	9 (9.57%)	1 (1.06%)	1 (1.06%)	0	6

MTBI = mild traumatic brain injury.

who state that the number is likely to be larger as many do not present to hospital (>600/100,000).

In Pakard's [1] review of the epidemiology he quotes the studies of Alves [33] and Evans [34] who reviewed the incidence of PTH in those with MTBI and quoted figures of between 30% and 50% as suffering with persistent symptoms. In Kraus et al. [6], a more recent controlled prospective study, the authors state that 79% of those suffering MTBI complained of persistent headache at 3 months. In our study, the figure is significantly lower than that reported by Kraus.

Further, the point prevalence of headache in the community needs to be distinguished as a baseline. Studies into PTH following MTBI need to establish that patients suffering headache following trauma can distinguish whether they are more severe than any of their preexisting headache patterns. In Rasmussen's [35] study of the general population in Denmark, he found that at any point in time 22% of women and 11% of men complain of headache from any cause. In Jensen et al.'s [36] 12-month retrospective study of PTH, it was suggested that 39.9% of patients had complained of headache prior to the injury and that the majority of those were women. This underlines the importance of determining whether headaches, post injury, are more severe or more common than those reported prior to the injury. In this study, the question was asked whether headaches suffered post-MTBI were worse than those (headaches) suffered before the injury. Data on pre-existing headache frequency was not collected in this study, and may have been useful in cross validating responses to this question.

However, it must be acknowledged that our study population had a predilection for male gender (77–78% male) and unusually has assault as the most common mechanism of injury. In

Jensen's [36] retrospective study of PTH following MTBI, it is indicated that more women report PTH than men.

In terms of the mechanism of injury, many of our subjects with MTBI were injured by assault and there were fewer falls or MVAs as mechanisms of injury. In Kraus's review [37], she indicated that the most frequent cause of all head injuries is MVA (42%), then falls (23%), then assaults (14%), and then sporting injuries (6%). This is confirmed by Povlishock [38]. The predominance of assault as a mechanism of injury in our study is likely to be related to the fact that St Vincent's Hospital is situated in the inner city of Sydney close to Kings Cross which is known as Sydney's red light district. Kings Cross also houses the only government sanctioned "safe injecting room" for intravenous drug users which is part of a trial to prevent overdose and death in Kings Cross's intravenous drug-using population. The location of hospital predisposes it toward managing a relatively high volume of victims of criminal assault or crimes related to alcohol or illicit drug use.

Interestingly, while patients suffering assault as related to a violent crime in Australia may pursue litigation, they do not fall under a mandatory third party insurance scheme, unlike those involved in motor accidents or accidents at work. Mickeviciene's study [7] suggests that the low incidence of postconcussive symptoms (11%) following MTBI, in Lithuania, is related to fact that that country lacks a mandatory third party insurance scheme for MVAs. This is emphasized by Carroll et al.'s 2004 WHO task force in MTBI [2] where she states that where symptoms persist, compensation and/or litigation are commonly a factor. It is interesting to note that in our study which found an incidence of 15.35% of PTH, most had suffered noncompensable injuries. Only 13% of the

Table 5 The table below indicates raw numbers and percentages in brackets

Scores	0 No Headache	1 as Before	2 Mild	3 Moderate	4 Severe	Lost to Follow-up
MTBI	58 (86.17%)	7 (8.97%)	3 (3.84%)	7 (8.97%)	2 (2.54%)	22
Minor injury controls	87 (97.75%)	0	2 (2.25%)	0	0	11

MTBI = mild traumatic brain injury.

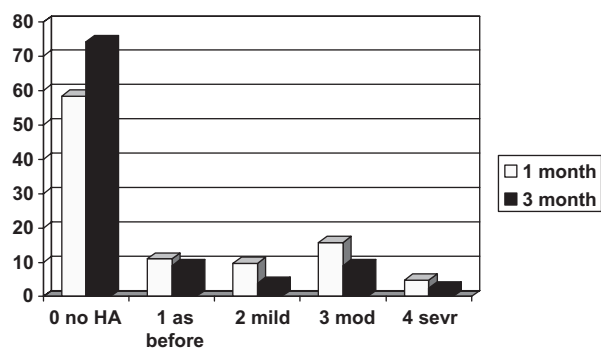


Figure 4 Natural history of PTH. X axis indicates the headache scores from the Rivermead Postconcussive Questionnaire. 0 = no headache, 1 = I do have a headache but it is no more of a problem now than before my injury, 2 = mild headache, 3 = moderate headache, 4 = severe headache. Y axis represents percentage of MTBI study population at 1 month (light color) and MTBI study population at 3 months (dark color). HA = headache; MTBI = mild traumatic brain injury; PTH = posttraumatic headache.

MTBI study population had sustained MVAs as a cause of their injury.

These three factors (predilection of males, assault as the most common mechanism of injury and far fewer patient's suffering compensable injuries) may offer some explanation for our study's lower than expected percentage of PTH at 3 months following injury.

In a number of studies of physician's attitudes to patients with MTBI and PTH, such as Evans et al. (1994) [21] and Warner et al. (1996) [22] many surveyed doctors felt that a number of trauma-related psychological factors may contribute to the incidence and persistence of PTH following MTBI such as psychological stress, anxiety, depression, and analgesic overuse. To examine the relative contribution of these factors to the incidence of PTH following MTBI, minor injury controls whose mechanism of injury was mostly low impact falls with presumed less perturbation of the resting posture of head and neck were utilized. It was assumed that they would be exposed to a similar degree of trauma-related anxiety, depression, psychological stress, and analgesic use, although no direct measures of these conditions

were taken. However, we found significantly less PTH in this minor injury control group. This may suggest that PTH following MTBI may be associated with organic causes rather than exclusively the psychosocial impact of injury.

In Packard's article [39] on the relationship of neck injury to PTH, he cites a number of studies which have hypothesized as to the causes of PTH. Many suggest that PTH is part of the whiplash complex and is generated from the rapid excursion of the neck through full range of movement while others suggest that PTH is more likely due to psychosocial factors such as distress or depression [2,8]. It must be noted that the mechanism of injury in MTBI can not be directly compared with that sustained through rapid acceleration-deceleration forces on the neck such as whiplash. Causes of headache have not been addressed in this article and speculation regarding possible application of these data to the whiplash group cannot be made.

In Barnat's [10] study of the demographic features of those with PTH, he noted that 46% were male with a mean age of 37 years with the females having a mean age of 38.4 years. Patients had an average of 12 years of education and most were skilled laborers. In total, 90% responded that they doubted that they had problems with drugs or alcohol.

Our study was not sufficiently powered to make a statistically meaningful analysis of the demographics of those with PTH. However, of those who had PTH at 3 months 45.5 % (five out of 11) had assault as mechanism of injury, 45.5% had falls (five out of 11), and 9 % (one out of 11) had MVAs as a mechanism of injury.

Povlishock [38] found that 64% of patients admitted following head trauma tested at or above the American legal limit for intoxication. In our study, we found that 20% of those with MTBI had breath alcohol levels above legal limits for driving in New South Wales (0.05 Blood Alcohol Level). These measures were recorded on average almost 14 hours after the injury. This compared with 8% of the minor injury group. The mean blood alcohol level in the MTBI group

Table 6 The table below indicates raw numbers and percentages in brackets

Scores of MTBI	0 No Headache	1 as Before	2 Mild	3 Moderate	4 Severe	Lost to Follow-up
1 month	48 (58.57%)	9 (10.97%)	8 (9.75%)	13 (15.85%)	4 (4.87%)	18
3 months	58 (86.17%)	7 (8.97%)	3 (3.84%)	7 (8.97%)	2 (2.54%)	22

MTBI = mild traumatic brain injury.

was 0.013 at the time of assessment (834 minutes after injury). It is likely that if breath or blood alcohol levels could have been taken at the time of admission to the ED that the number found to have readings above 0.05 may have been higher than 20%. While, one might suggest that those patients intoxicated at the time of the injury may have more severe injuries or may not be routinely offered analgesia due to the risk of drug interactions, the association of alcohol at the time of injury and the risk of developing PTH has not been established. Patterns of alcohol consumption are likely to play a role in the management of chronic PTH as many analgesic medications interact with alcohol.

There were a number of methodological features that may have diminished the quality of the study such as the fact that patients were asked about headache as part of the Rivermead Post Concussive Questionnaire at the 1 and 3 months post injury, without reference to a VAS of pain. Our study also had an atypical spread of mechanism of the MTBI with a predilection for assault as well as a largely male sample. This may have negatively skewed our results. The fact that few of the subjects with minor head injuries had compensable injuries through a mandatory third party or workers compensation system, may have further skewed the figures although some authors may view this as a characteristic of the population that contributes to the accuracy of the result. Notwithstanding, the prevalence of PTH at 3 months following MTBI in our prospective controlled study was 15.35% which is a significant finding when compared with our minor injury control group prevalence of 2.25% ($P \leq 0.001$). This result adds to the evidence that a significant proportion of patient who suffer MTBI will complain of PTH at 3 months.

Conclusion

To the authors' knowledge, this represents the first Australian prospective controlled study in the development of PTH following MTBI. In this study, we found that 15.35% of those who suffered an MTBI complained of PTH at 3 months. It would benefit health planners, administrators, and service providers to undertake a larger multicentered prospective trial of the incidence of PTH following MTBI with a 12-month follow-up to examine prognostic factors, service utilization, and possible preventative interventions.

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